

SSH Cheat Sheet

This is a document prepared to help candidates unfamiliar with SSH in connecting to a virtual machine/server over SSH. Based on your system's operating system, you can refer to the right notes below:

Common notes: `ssh_port` is 22

Common tools to edit files on the server: `nano`, `vim`

If you write code on your laptop, you can use `scp` to copy files to the server to run them.

- **Windows:**

- If you have WSL (Windows Subsystem for Linux) (<https://docs.microsoft.com/en-us/windows/wsl/install>) enabled, skip the Windows section and just use the Linux section after opening a WSL terminal window.
- If your version of windows has the "ssh" utility preinstalled, skip the Windows section and just use the Linux section after opening the terminal. You can check by opening Windows Powershell and running the command `'ssh -V'`. If it returns the version of OpenSSH, you can follow the Linux section instead of installing PuTTY.

If the above two points don't apply to you:

- Install PuTTY from here: <https://www.putty.org/>
- This is an ssh client. Just start this up, enter your host IP, username and port. Choose the connection type as ssh.
- Guide here: <https://www.ssh.com/ssh/putty/windows/>
- When asked for the password, enter your password into the dialog box and you should get a bash terminal. We suggest using copy-paste. You should be able to paste into the terminal with `Ctrl+Shift+V` or a single right click.
- Remember, the password input won't be visible on the PuTTY terminal so whether you type manually blindly or paste from clipboard, make sure your PuTTY window is in focus and that you are not double-pasting or mistyping, and press Enter once you have input the password.
- You can try pasting the password in a normal terminal just to confirm you have copied it correctly into your clipboard.

- **Linux:**

- OpenSSH comes packaged with most distros. Open up a terminal and type in `'ssh'` to see if it works.
- If it doesn't, install it using your package manager. A few examples (use the one that corresponds to your operating system/package manager):

```
sudo apt install openssh-client
sudo pacman -S openssh
sudo dnf install -y openssh-clients
```
- Guide to install openssh on linux:
<https://averagelinuxuser.com/how-to-install-and-use-ssh-on-linux/>

- Once the installation is done, just open up a terminal and type in:
`ssh -l <ssh_username> -p <ssh_port> <ip_address>`
- You'll be prompted to enter the password. Enter your password and you should get a bash terminal. Remember, the password input won't be visible on the terminal so whether you type manually blindly or paste from clipboard, make sure your terminal window is in focus and that you are not double-pasting or mistyping, and press Enter once you have input the password.

- **MacOS:**

- Most modern macOSes have an ssh client built in
- Verify by running a terminal and typing in ssh to see if it works
- If it doesn't, for some reason, ssh can be installed via brew
- Install brew as shown here: <https://docs.brew.sh/Installation>
- Once that is done, just type the following into a terminal:
- `brew install openssh`
- Once the installation is done, just open up a terminal and type in:
`ssh -l <ssh_username> -p <ssh_port> <ip_address>`
- You'll be prompted to enter the password. Enter your password and you should get a bash terminal.